

Baseline → Multi-Task MLP: Risks, Benefits & Pro

Emergency Department Decision Support

Baseline: gate_first_nomlp_baseline • 2025-08-23

Author: Solo maintainer (ED physician)
Collaboration: LLM-assisted • Open-source / non-profit

Executive Summary

Why change: Gate_pos=admit is confounded by operational, social, and policy factors. We keep gate-first as the arbiter and add a multi-task MLP to learn non-linear clinical dynamics.

- Benefits: interpretable clinical heads, actionability, non-linear learning, robustness to ops drift via gate floors.
- Risks: label confounding or leakage, ops non-stationarity, class imbalance, negative transfer across tasks, silent degradation.
- Guardrails: policy-robust labels, temporal validation, calibration, small/shared trunk with regularization, gate-first priority floors, simple acceptance criteria.

Benefits (Clinical • Technical • Operational)

- Clinical: interpretable targets (complexity, resource intensity, deterioration risk, readiness, completion).
- Technical: multi-task representation shares signal across heads; captures interactions beyond calculators.
- Operational: improves sorting/throughput without predicting beds; integrates with existing calculators and gates.

Risks & Failure Modes

- Label confounding: targets accidentally encode bed manager or consult culture.
- Label leakage: using post-t results/actions in features; identity mapping from calculators to labels.
- Class imbalance: few true deteriorations → unstable thresholds and poor calibration.
- Non-stationarity: seasonality, protocol changes, lab turnaround drift.
- Negative transfer: one task harming another in multi-task training.
- Silent degradation: performance fades after threshold tweaks or data schema changes.

Mitigations (Keep it Sound)

- Policy-robust label design: use future clinical events and protocolized prerequisites; exclude bed/insurance orders.
- Temporal CV: split by week/shift; report per-slice (day/night, weekday/weekend).
- Calibration: temperature/Platt for probabilities; isotonic for ordinal cutpoints; monitor reliability by group.
- Regularization: small trunk (2-3 layers), dropout, weight decay, early stopping on temporal-val.
- Loss balancing: uncertainty weighting / DWA if tasks conflict; per-head masking for missing labels.
- Gate-first floor: gates set a minimum operational priority; MLP can only raise/refine, never veto.
- Change control: one change per session; run sanity pack; tag baseline if green.

Targets & Label Rubrics (Concise)

- Clinical Complexity (0-10): scaled calculators at t + critical clinical events in $t \rightarrow t+2h$; normalize, clamp.
- Resource Intensity (low/med/high): therapy intensity over $t \rightarrow t+4h$ (O2 tier, vasopressor, continuous monitoring, serial labs). Avoid consult-count and paperwork proxies.
- Deterioration Risk (0-1): code, new pressor, HFNC/NIV, critical BGA, $GCS \downarrow \geq 2$, $qSOFA \uparrow \geq 2$ in $t \rightarrow t+4h$; handle imbalance.
- Time to Clinical Readiness (ordinal): last required result/consult/imaging complete + stable 30-60 min; avoid 'admit order'.
- Workup Completion (0-1): completed_required/required from protocolized plan at/just before t (gate engine), not ad-hoc orders.

On-Track Checklist (Per Session)

- Repro path: CONFIG → G1 → G2 → (HL7/Calcs/Bundle if needed) → Triage + Sanity: TTL_MAX_OK, GATES_WIRED_OK, BUNDLE_SMOKE_OK.
- Data snapshots: features $\leq t$; labels from future windows; censor appropriately; no ops features.
- Temporal split: train/val/test by time; check day/night and weekday/weekend slices.
- Calibration: reliability curves; group-wise (sex/age) checks; recalibrate if drifted.
- Ablation: without calculators; ensure performance degrades gracefully (not identity mapping).
- Gate integration: priority = max(gate_floor, f(MLP)); verify no gate suppression.
- Release note: what/why/validation (≤ 3 bullets); if green, tag baseline; else revert.

What to Avoid (Anti-Patterns)

- Predicting admission or bed assignment.
- Counting consults or orders as resource intensity (policy-driven).
- Using future results/actions as features at t (leakage).
- Training/evaluating with random splits across time.
- Letting MLP veto gates or lower alert priority.
- Uncalibrated probabilities on rare outcomes.
- Multiple edits per session without re-running sanity pack.

Acceptance Criteria (Go/No-Go)

- Temporal-CV: deterioration AUCPR stable across shifts/weekday vs weekend ($\pm 10\%$).
- Calibration: reliability within $\pm 5\%$ for deterioration overall and by groups (post-calibration).
- Multi-task $\geq$ single-task on at least one head without degrading others (paired test).
- Gate-first contract: no safety gate suppression; improved recall@K for who needs attention next.
- Sanity pack all green; baseline manifest updated (thresholds + function fingerprints).

Roadmap (Minimal, When Ready)

- Phase 1: Log label rubrics and counters; dry-run heads without affecting priority.
- Phase 2: Calibrate and plug heads into 'operational priority' as refiners only.
- Phase 3: Monitor drift; periodic recalibration; optional hazard head for time-to-readiness if censored labels grow.

Appendix: Snapshot Protocol

- Create snapshots at triage and every 15–30 min.
- Compute calculators at t ; build labels only from $t \rightarrow \text{horizon}$.
- Mask missing labels per head; multi-task handles per-example masking.
- Exclude ops features (beds, transfer orders, insurance). Keep clinical signals and standardized protocol artifacts.